Report: Sanskrit to English Neural Machine Translation
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1. Objective

The aim of this project was to build a Neural Machine Translation (NMT) model capable of translating Sanskrit sentences to English using a sequence-to-sequence deep learning system with PyTorch. It was trained on 10K pairs of parallel sentences in Sanskrit/English language and tested on an independent validation dataset.

2. Dataset

Dataset	Size
Training	10,000 sentence pairs
Validation	1,000 sentence pairs
Test	1,000 sentence pairs

Language	Vocabulary Size
Sanskrit	32,598
English	11,984

Maximum token IDs were verified to match the vocabulary size, confirming that the vocabulary construction and tokenization were correct

Preprocessing

- Unicode normalization
- Word tokenization
- Vocabulary construction
- Padding
- Special tokens (<PAD>, <SOS>, <EOS>, <UNK>)

3. Model Architecture

The implemented Neural Machine Translation model consists of:

- Encoder-Decoder architecture
- Embedding layers
- Attention mechanism (if implemented)
- Teacher forcing during training
- Cross-Entropy Loss
- Adam optimizer
- CUDA acceleration

Model statistics:

Property	Value
Trainable Parameters	41,565,905
Device	NVIDIA GPU (CUDA)

4. Training Configuration

Parameter	**Value**
Epochs	**25**
Optimizer	**Adam**
Loss Function	**Cross Entropy**
Device	**CUDA GPU**

5. **Training Results**

Epoch	Train Loss	Validation Loss
1	6.3409	6.2877
2	5.7005	6.1743
3	5.3194	6.2489
4	4.9841	6.0993
5	4.6538	6.1944
10	3.3640	6.5092
15	2.9610	6.5761
20	2.8609	6.6015
25	2.8599	6.6240

Best validation loss:
6.0993 (Epoch 4)

6. **Evaluation Metrics**

Metric	Value
BLEU Score	0.0185
BERT Precision	0.8149
BERT Recall	0.8330
BERT F1	0.8234
Token Accuracy	12.30%
Average Validation Loss	6.0945

7. **Inference Performance**

Metric	Value
Test Sentences	1000
Total Time	268.33 sec
Average Time/Sentence	0.268 sec

The model successfully generated predictions for all test sentences and produced the required submission.csv.

8. Sample Predictions

Example 1**Input (Sanskrit)**

एक्लिप्स् इति प्रोग्रामर् कृते दोषान्वेषणे अपि साहाय्यं करोति।

Prediction In this tutorial, we will be used in the .

9. Analysis

The model learned the simple structure of English sentences and produced grammatically plausible sentences. However, in general, the translations were not related to the Sanskrit source. There are a number of observations that can be made: Training loss continuously decreased, indicating effective optimization on the training data. The validation loss did not get better until the Epoch 4 and worsened from that point on, suggesting overfitting. The lexical similarity to reference translations is low, as shown by the low BLEU score (0.0185). There is a relatively high BERTScore F1 (0.8234) although there is a low number of exact words matched. The token accuracy score of 12.3% shows that many tokens have been predicted without matching any of the reference translations. The Sanskrit input was exactly the same, with common English sentence patterns like "Now" and "So" and "click on this file" and "tutorial" appearing in the generated translations, indicating that the model was learning the common English sentence structures, but the semantics of the Sanskrit input were not captured.

10. Reasons for Low Performance

The observed performance was the result of several factors:

- The training corpus used is quite small with only 10,000 sentence pairs for training a neural machine translation model from scratch.
- The model has more than 41 million trainable parameters, which means it can be susceptible to overfitting the model to a relatively small amount of data.
- The word level tokenization generates a big vocabulary (32,598 Sanskrit tokens, 11,984 English tokens), resulting in a large number of rare words.
- Sanskrit language has a very rich morphology, which is why there are sparse word distributions in Sanskrit.
- No pretrained multilingual encoder or decoder was used.

11. Possible Improvements

Future work will enhance the quality of the translation by:

- Training more than 100,000 pairs of sentences.
- Make use of subword tokenization (such as SentencePiece or BPE). Using pretrained multilingual models like mBART, mT5 or NLLB.
- Using label smoothing and dropout.
- Instead of greedy decoding use beam search decoding.
- Including learning rate scheduling and early stopping.
- Training for more epochs using better regularization.

The Neural Machine Translation (NMT) system from Sanskrit to English was successfully developed, trained, evaluated and deployed for inference. The model was correctly implemented, learned the sentence generation and output fluent English, but translation quality was limited because of the small training set and the problem of overfitting of the model. The best model was obtained at Epoch 4 with a validation loss of 6.0993. Despite the low BLEU score, the project provides an end-to-end example of developing, training, testing, and deploying a neural machine translation system. Translation accuracy will be greatly enhanced by larger data sets, subword tokenization, and pretrained multilingual models in the future.

References

1. Bahdanau D., Cho K., Bengio Y., *Neural Machine Translation by Jointly Learning to Align and Translate*, ICLR 2015.
2. Sutskever I., Vinyals O., Le Q., *Sequence to Sequence Learning with Neural Networks*, NeurIPS 2014.
3. Papineni K. et al., *BLEU: A Method for Automatic Evaluation of Machine Translation*, ACL 2002.
4. Zhang T. et al., *BERTScore: Evaluating Text Generation with BERT*, ICLR 2020.
5. PyTorch Documentation.
6. Hugging Face Transformers Documentation.